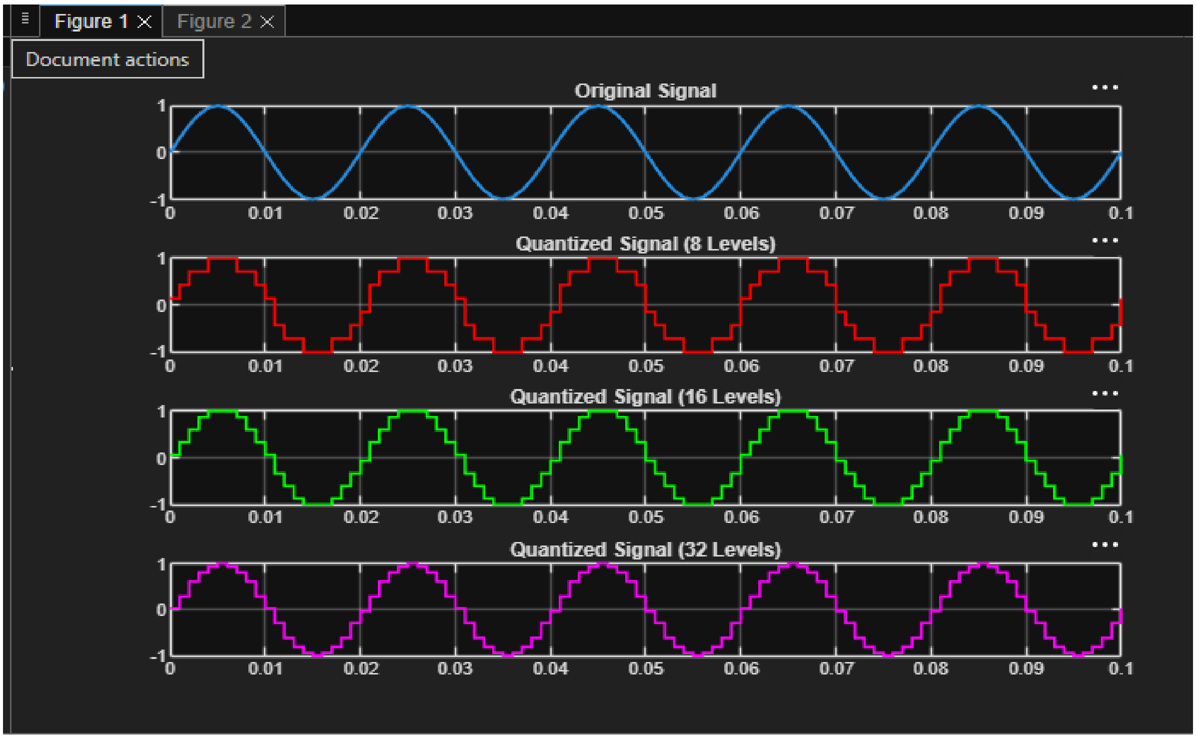
Experiment 6.1: Uniform Quantization in MATLAB

Aim: To implement uniform quantization for a sinusoidal signal using 8, 16 and 32 quantization levels and compare the quantization error.

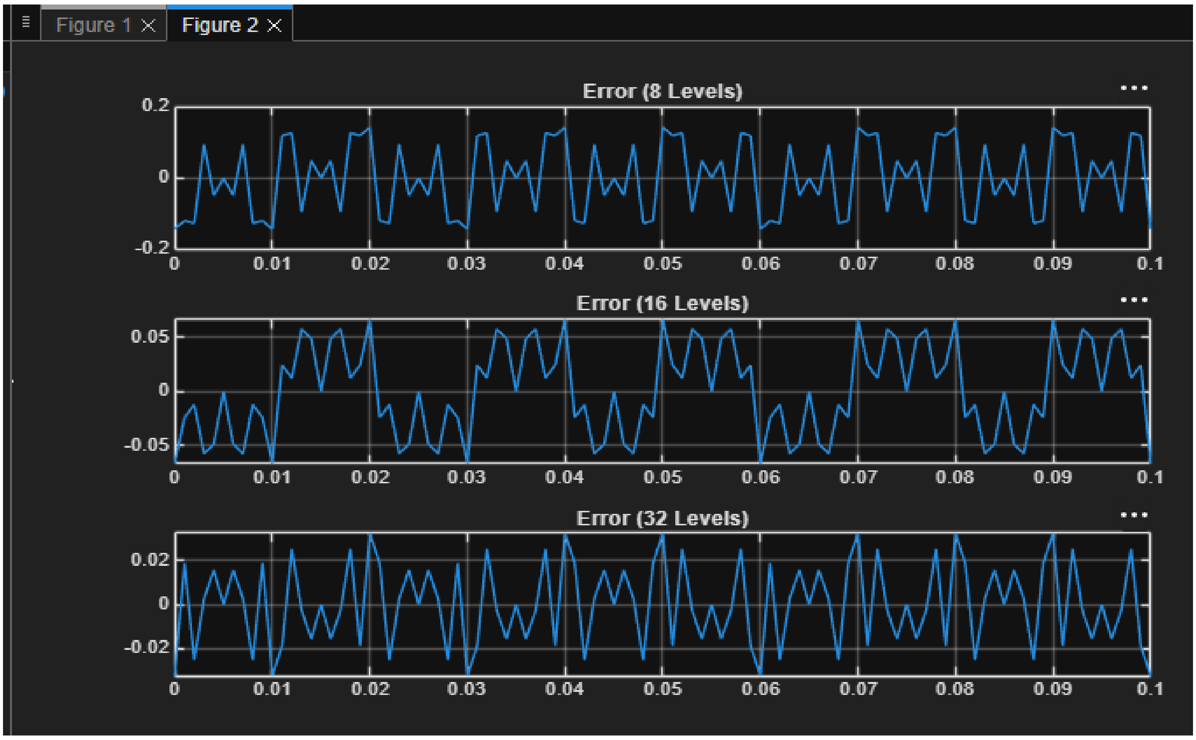
MATLAB Code

```
clc;
clear;
close all;
%% Signal
fs = 1000; f = 50;
t = 0:1/fs:0.1;
x = sin(2*pi*f*t);
%% Quantization Levels
L = [8 16 32];
colors = {'r','g','m'};
figure;
for i = 1:length(L)
    % Quantization
    xq{i} = round(((x + 1) * (L(i) - 1) / 2)) * (2 / (L(i) - 1)) - 1;

    % Error & MSE
    e{i} = x - xq{i};
    mse(i) = mean(e{i}.^2);

    % Plot Quantized Signals
    subplot(4,1,i+1);
    stairs(t,xq{i},colors{i},'LineWidth',1.2);
    title(['Quantized Signal (' num2str(L(i)) ' Levels)']);
    grid on;
end
% Original Signal
subplot(4,1,1);
plot(t,x,'LineWidth',1.5);
title('Original Signal'); grid on;
% Display MSE
fprintf('MSE for 8 Levels = %f\n', mse(1));
fprintf('MSE for 16 Levels = %f\n', mse(2));
fprintf('MSE for 32 Levels = %f\n', mse(3));
%% Error Plots
figure;
for i = 1:3
    subplot(3,1,i);
    plot(t,e{i});
    title(['Error (' num2str(L(i)) ' Levels)']);
    grid on;
end
```

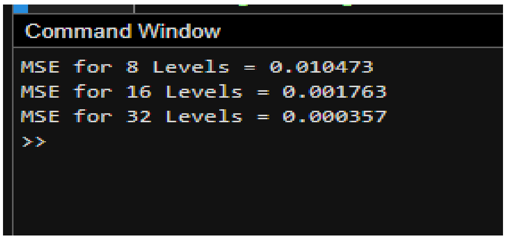
Output - Quantized Signals



Output - Error Plots



Command Window Output



Result: Uniform quantization was implemented successfully. As the number of quantization levels increased, the Mean Square Error decreased.